

Revised Generation Interconnection System Impact Study Report

For Queue Project Q19-1244

Mandan Junction 138 kV
August 03, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1244. The Interconnection Customer proposes a 101.5 MW solar photovoltaic facility interconnecting at the Mandan Junction 138 kV substation in the Pennsylvania area. The request was received on July 11, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 466 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$732,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: SARGEN CANYON to HEMET 138.0 kV
Contingency: SMITH to HEMET 345.0 kV
Loading: 113.7% (Rating: 203 MVA)
DFAX: 19.2%
Pre-project loading: 104.1%
2. Facility: CANYONVILLE to SARGENT CANYON 138.0 kV
Contingency: SMITH to HEMET 345.0 kV
Loading: 108.5% (Rating: 225 MVA)
DFAX: 19.2%
Pre-project loading: 99.8%
3. Facility: CASCADE VLY to CANYONVILLE 138.0 kV
Contingency: SMITH to HEMET 345.0 kV L
Loading: 107.9% (Rating: 236 MVA)
DFAX: 19.2%
Pre-project loading: 99.6%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SARGEN CANYON to HEMET 138.0 kV	Rebuild line with higher-capacity conductor	\$2,421,000	15% (\$353,000)
CANYONVILLE to SARGENT CANYON 138.0 kV	Reconductor line	\$1,531,000	15% (\$223,000)
CASCADE VLY to CANYONVILLE 138.0 kV	Reconductor line	\$1,068,000	15% (\$156,000)
	Total allocated to project		\$732,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.